Unit 5, Lesson 6: Currency Conversions

Benchmark

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Learning Target

☐ I can convert between two different forms of currency.

5.6.1 Warm-Up: Would you Rather?

Consider the exchange rates from currency in a Mathlandia country, Scalenia, compared to the United States (U.S.) dollar.

	U.S. dollar ($) Equivalent	**Currency per U.S. dollar ($)**
Scalenia 1 sal (§)	1.5939	0.6274

1. Would you rather have §100 or $100? Explain your reasoning.

5.6.2 Collaboration: Reasoning and Sense Making

Work with your partner on the following problems.

1. The exchange rate used to change Multiplicity macs (ɱ) to Proportia pops (ᑭ) is $\frac{5}{8}$. Complete the statement.

   For every _____ mac(s) there are ______ pop(s).

2. On October 28, 2020, Lorenzo's family exchanged 250 U.S. dollars for $5{,}330$ Mexican pesos.

U.S. dollars	Mexican pesos
250	$5{,}330$
25	
2.50	

   a. Complete the table.

   b. Complete the rate to represent the exchange rate of Mexican pesos per U.S. dollar on October 28, 2020.

$$\frac{\quad}{1\ US\ dollar}$$

   c. Lorenzo bought a blanket for 247 pesos that day. Without calculating, estimate the cost of the blanket in dollars.

   d. What was the cost of the blanket in dollars?

   e. What was the exchange rate in terms of U.S. dollars per peso on October 28, 2020? Write your answer as a decimal to the nearest hundredth.

   ________ U.S. dollars per Mexican peso

5.6.3 Guided Instruction: Converting Currencies

An **exchange rate** is a ratio that compares two different currencies. An exchange rate can be written as a fraction, ratio, or decimal.

When the exchange rate is written as a unit rate, it shows the value of a certain currency for each one unit of the other currency.

Ratio tables, proportions, or other equations are useful when converting one currency to another.

Recall the exchange rates from the Mathlandia Exploration.

Equivaland 1 equi, ∈	≈	1.32	sals, §
		0.85	pop, ℘
		15.90	peaks, Λ̊
		0.30	var, ṽ
		7.75	ops, ⊕

1. Determine the values for each exchange.

   a. 45 equis = _________ pops

   b. $1{,}113$ peaks = ________ equis

2. Emma travels to Scalenia from Equivaland. She exchanges 200 equis. How much does she receive in sals?

3. Emma purchases these items while in Scalenia. What is the cost of each item in equis, ∈?

   a. A necklace for §64.35

   b. A scarf for §11.22

5.6.4 Guided Practice: Converting Currencies

1. Jaxon exchanged 500 dollars for 420 euros.

    a. What was the exchange rate in euros per dollar?

    b. Jaxon paid 16.49 euros for a shirt. Do you expect the price of the shirt to be more or less than 16.49 dollars? Explain your thinking without doing any calculations.

    c. What is the price of the shirt in dollars?

2. Keisha travels from Proportia to Variableya. The currency exchange rate is 1 pop = 0.35 var. Keisha exchanges 120 pops. How much does she receive in vars?

5.6.5 Your Turn!

Recall the exchange rates from the Mathlandia Exploration.

Scalenia 1 sal, §	≈	0.64	pop, ℘
		12.05	peaks, Å
		0.76	equi, ∈
		0.23	var, ṽ
		5.88	ops, ⊕

1. Determine the equivalent values for each currency exchange.

   61 sals = _______ vars   _______ sals = 106.40 equis

   18 peaks = _______ sals   90 sals = _______ ops

2. Logan lives in Equivaland and wants to take his family on vacation.

   The exchange rate is 1 equi (∈) = 7.75 ops (⊕).

   Logan has saved up 400 equis for the trip. Aunt Emma, from Prismopolis, sent 1,200 peaks to help pay for the trip. The exchange rate is 1 equi = 15.9 peaks.

   With Aunt Emma's gift, how much does Logan have available for the trip in equis? Round to the nearest hundredth.

5.6.6 Wrap-Up: Ticket Out the Door

1. Milo exchanged $\$400$ for 367 francs. Complete the two statements based on this currency exchange.

   1 dollar = ___________ franc(s)

   1 franc = ___________ dollar(s)

2. The exchange rate from Integerina ops to Prismopolis peaks is 1 op $=$ 2.05 peaks.

   Ava exchanged 500 peaks for ops. How many ops did she receive?